CYBERSECURITY PRACTICES AND DATA SECURITY POLICY:

AN IT PROFESSIONAL PERSPECTIVE

Prepared by:

Dulay, Kyle M.

Pangilinan, Aris Guiller S.

BS Information Technology – 4th Year

Subject: Data Privacy and Security

Instructor: Mr. Wendyll Ojasto

Date: October 2026

1. INTRODUCTION

Background of the Topic

Cybersecurity and data security are now priorities for modern organizations that rely heavily on IT infrastructure. Digital systems help daily operations run smoothly. They also put enterprises at risk from serious threats, such as unauthorized access, phishing schemes, malware and data breaches.

To reduce these risks. Keep operations steady organizations use strong cybersecurity practices and strict data security policies. These frameworks work together to protect networks, keep digital assets safe and protect system users throughout the organization.

This term paper looks at how cybersecurity measures and data policies are used in real-world enterprise settings. By mixing research with insights, from an active IT professional this study checks how frameworks work in practice, finds common bottlenecks and suggests ways to improve overall security level.

Purpose of the Paper

The main goal of this term paper is to look at how cybersecurity practices and data security policies are used in a working IT environment. The paper uses both a review of existing research and a detailed interview with an IT professional. This approach helps to see how the basic cybersecurity ideas

learned during the Bachelor of Science in Information Technology (BSIT) program are put into action, in real-world jobs.

Specifically this paper wants to find out what cybersecurity methods the organization uses. It also wants to examine the data security policies they follow. The paper looks at the problems that come up in operations. Based on this it gives suggestions to make security stronger and improve compliance.

Objectives

This term paper aims to achieve the following specific objectives:

1. Identify the core cybersecurity practices and tools deployed within an operational IT environment.
2. Examine the specific data security policies, administrative procedures, and access controls enforced by IT personnel.
3. Analyze the operational challenges and threat vectors encountered by IT professionals in the workplace.
4. Correlate workplace security measures with established industry policies, standards, and academic literature.
5. Evaluate the role of employee security awareness training and organizational governance in mitigating security risks.
6. Formulate actionable, practical recommendations to enhance organizational cybersecurity and data security management.

Scope of the Paper

This paper looks into how organizations put cybersecurity measures in place and make sure data security policies are followed. The scope covers topics such as network defense, data lifecycle protection, access control models, identity verification, administrative policy enforcement, security awareness, operational challenges and strategic security enhancements. These topics come from research literature and from the feedback of an IT professional.

To keep confidentiality, privacy and non-disclosure agreements this paper does not gather, keep or share sensitive organizational data. Sensitive data includes network configurations, system passwords, user credentials, proprietary source code and internal topology diagrams. All information that the interviewee gives is kept to non-proprietary operational insights that fit academic analysis. This keeps the research sound and protects the organization’s security posture.

2. REVIEW OF RELATED INFORMATION

Cybersecurity Practices

Modern cybersecurity uses a Defense-in-Depth approach. Defense-, in-Depth approach combines measures, technical measures and administrative measures. Layered approach protects information and systems. Layered approach ensures that if one security measure fails other security measures remain active to stop access.

- Network Architecture & Segmentation: Organizations isolate internal environments through IP address range partitioning and Virtual Local Area Networks (VLANs). Subnetting prevents unauthorized lateral movement, restricting lateral threat propagation if a single subnet is compromised.
- Identity and Access Management (IAM): Adhering to the Principle of Least Privilege (PoLP) and Role-Based Access Control (RBAC) ensures users and automated processes are granted only the network rights required to perform their explicit functions. Identity verification is fortified using Multi-Factor Authentication (MFA).
- Endpoint & DNS Security: Enterprise networks employ centralized Endpoint Detection and Response (EDR) software alongside custom Domain Name System (DNS) filtering configurations. Secure DNS routing blocks domain lookup requests directed toward known malicious, phishing, or command-and-control (C2) domains.
- Vulnerability & Threat Management: Proactive vulnerability assessment schedules, systematic software patch management, and stateful Intrusion Detection/Prevention Systems (IDS/IPS) continuously inspect incoming and outgoing traffic for anomalous behaviors.

Data Security

Data security aims to protect information by ensuring that it remains confidential, accurate, and accessible when needed. These protections apply

throughout the data lifecycle, including when data is stored, transmitted over a network, or being processed.

1. Confidentiality & Data Encryption: Data must remain inaccessible to unauthorized entities. Organizations apply cryptographic standards such as Advanced Encryption Standard with 256-bit keys (AES-256) for data at rest (database storage, file systems) and Transport Layer Security (TLS 1.3) or Internet Protocol Security (IPSec) tunnels for data in transit.
2. Integrity Protection: Cryptographic hashing functions (such as SHA-256) and digital signatures verify that system files and transmitted database records have not undergone unauthorized modification, corruption, or tampering.
3. Availability & Business Continuity: Systems enforce continuous uptime through redundant storage architectures and strict adherence to the 3-2-1 Backup Strategy (three copy backups across two different media types, with at least one copy held offsite or in a secure cloud repository).

Data Security Policies

A Data Security Policy is a set of rules and guidelines that explains how an organization should protect, manage, and use its information. It helps employees understand their responsibilities when handling company data and

provides guidance on maintaining security and following organizational requirements.

- **Acceptable Use Policy (AUP):** This policy explains how employees should properly use company computers, networks, internet access, and email. It may also include guidelines for using personal devices for work, helping ensure that employees use company resources responsibly and securely.

- **Data Classification Policy:** This policy organizes information according to its level of sensitivity, such as public, internal, confidential, or restricted. It helps determine how different types of information should be accessed, stored, shared, retained, and protected.

- **Incident Response Plan (IRP):** This plan outlines the steps an organization should follow when a security incident occurs, such as a data breach or ransomware attack. It explains how to identify the problem, limit its impact, remove the threat, and restore affected systems. It also identifies the responsibilities of IT personnel during the response

3. IT PROFESSIONAL INTERVIEW AND DISCUSSION

To see how theoretical concepts function in everyday settings, this study gathers firsthand insights from an IT professional managing live network

infrastructure and technical event operations.

3.1 Interviewee Profile

- **Full Name:** Franklin Opinion
- **Current Position / Job Role:** LiveProd Coordinator & Event Technical Coordinator
- **Area of Specialization:** Event Technical Coordination & IT Infrastructure Management
- **Years of Experience:** 10 Years

As a LiveProd Coordinator and Event Technical Coordinator Mr. Opinion handles temporary and permanent network infrastructure in live production settings. The role of LiveProd Coordinator and Event Technical Coordinator demands that Mr. Opinion manage network traffic, protect live media feeds, separate connected devices and keep the system running at 100 percent uptime while safeguarding network security limits.

3.2 Interview Findings

The results from the interview are listed below under four areas: general cybersecurity practices, data security measures, policy enforcement and technical challenges.

Cybersecurity Practices

According to Franklin Opinion, the organization implements several multi-layered cybersecurity practices to protect its systems and information from cyber threats.

One of the primary practices mentioned was the implementation of multi-layer network security and IP address segmentation using separate IP address ranges to isolate network segments. Trusted devices are also grouped together to facilitate access management and strengthen overall network protection.

The interviewee also explained that DNS security configurations are used to shield users and devices from potential security threats posed by malicious or suspicious websites.

Data Security Practices and Policies

The interviewee stated that sensitive information is protected from unauthorized access through structured network controls.

Virtual Local Area Networks (VLANs) are utilized to separate user devices, ensuring that internal production networks remain logically isolated. For guest users and transient connections, dynamic IP address allocation is enforced to support network isolation and simplify IP address management.

In terms of data security policies, the organization configures firewall rules to restrict unauthorized access to shared files and network

resources. Additionally, Virtual Private Networks (VPNs) are required for remote operations to mask actual IP addresses and securely route encrypted network traffic through external servers, safeguarding data during transmission.

Cybersecurity Challenges

According to the interviewee, one of the primary operational challenges faced in IT infrastructure management is maintaining security across complex, high-density environments where numerous untrusted devices connect to the network.

The interviewee also identified the continuous risk of network interception, unauthorized access, and malicious domain threats as ongoing challenges, emphasizing that security configurations—such as firewall rules and DNS filtering—must be constantly updated to protect sensitive data and operational feeds.

3.3 Discussion

The interview results are mostly in line with the cybersecurity methods and data security ideas that were talked about in the review of information.

For example the use of VLAN segmentation IP address separation and firewall rules that were found during the interview shows the idea of keeping access limited to people who are allowed and stopping people who are not allowed from moving around the network. Also the way the organization uses

DNS setups, dynamic IP assignments and encrypted VPN connections shows how technical measures are used in real life to keep data private and safe.

The interview also shows how important it is to manage infrastructure in a thinking way when it comes to cybersecurity. Even though the rules talk a lot about identity and access policies, actual IT work depends a lot on automated network controls to make sure that internal parts of the system stay safe when devices that are not trusted are connected to the network.

The results show that good cybersecurity is about using layers of technical protections making sure network policies are followed strictly and always working to stop threats.

4. CONCLUSION AND RECOMMENDATIONS

4.1 Conclusion

This term paper examined how cybersecurity practices and data security policies help organizations protect their systems, networks, and information. It discussed related studies, recognized standards such as the NIST Cybersecurity Framework (CSF) 2.0 and ISO/IEC 27001, the Philippine Data Privacy Act of 2012 (Republic Act No. 10173), and the information gathered from an interview with Mr. Franklin Opinion.

Based on the findings, cybersecurity requires continuous attention and regular monitoring rather than relying only on security software. Network

segmentation, VLANs, IP address management, and firewall rules can help control network access and reduce the risk of unauthorized connections. The use of VPNs and secure DNS configurations can also help protect data and network communications from potential threats.

The findings also show that technology and organizational policies should work together to maintain security. Clear rules, proper network configurations, and appropriate access controls can help organizations protect sensitive information and reduce cybersecurity risks. Overall, maintaining a secure IT environment requires regular improvements, proper security practices, and the responsible handling of data.

4.2 Recommendations

Based on the research findings and discussion, the following actionable recommendations are proposed:

- Implement Automated Network Access Control (802.1X NAC): Standardize device authentication by deploying IEEE 802.1X port-based Network Access Control to validate hardware identity before granting network access.
- Deploy Encrypted DNS Protocols: Upgrade standard DNS configurations to DNS-over-HTTPS (DoH) or DNS-over-TLS (DoT) to prevent DNS eavesdropping and spoofing attacks.

- Transition Toward Zero Trust Network Architecture (ZTNA): Replace traditional perimeter VPN access with Zero Trust Network Access (ZTNA) solutions, granting application-specific privileges based on verified device identity and context.
- Conduct Regular Automated Penetration Audits: Supplement manual firewall rule reviews with quarterly vulnerability scanning and automated configuration audits to detect misconfigurations before deployment.
- Establish Incident Response Playbooks for Infrastructure Teams: Develop documented incident containment playbooks tailored specifically for live production and guest network environments.

REFERENCES

Information Technology Laboratory (National Institute of Standards and Technology). (2024). NIST Cybersecurity Framework (CSF) 2.0 (NIST Special Publication 1299). National Institute of Standards and Technology. https://doi.org/10.6028/NIST.SP.1299

International Organization for Standardization. (2022). Information security, cybersecurity and privacy protection — Information security management systems — Requirements (ISO/IEC Standard No. 27001:2022). https://www.iso.org/standard/82875.html

Republic of the Philippines. (2012). Republic Act No. 10173: An Act protecting individual personal information in information and communications systems in the government and the private sector (Data Privacy Act of 2012). Official Gazette of the Republic of the Philippines. https://www.officialgazette.gov.ph/2012/08/15/republic-act-no-10173/

Stallings, W. (2021). Cryptography and Network Security: Principles and Practice (8th ed.). Pearson.

APPENDIX A

1. What is your full name, current job role/title, and field of IT specialization?
2. How many years of professional experience do you have in the IT industry?
3. What common cybersecurity practices do you use to protect computer systems, infrastructure, and digital assets from cyber threats?
4. How are IP addresses, DNS settings, and device groups managed to ensure network protection?
5. How do you protect sensitive information and prevent unauthorized individuals from accessing corporate resources?
6. How are dynamic guest users and transient devices handled to maintain overall data security?
7. What specific firewall configurations or VPN controls are implemented to enforce data security policies?

8. What common cybersecurity problems or operational challenges do you encounter in your daily work, and what do you recommend to improve data security?

APPENDIX B

INTERVIEW DOCUMENTATION

In accordance with research ethics, non-disclosure agreements and privacy guidelines established by the interviewee's organization it was not allowed to take direct visual documentation during the interview. This includes photographs, video recordings or screen captures.

To ensure compliance with data privacy standards and to protect sensitive information related to organizational security posture the qualitative data for this study was collected using a structured direct questionnaire. The data was gathered through a verified discussion, with Mr. Franklin Opinion, who serves as the LiveProd Coordinator and Event Technical Coordinator.